

TPC-MVP3: A Typed Route Decision after TPC-121–131: Executable Archives, the Arithmetic Frontier, and Certified Branching

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Abstract

TPC-MVP2 separated exact interfaces from open arithmetic inputs but returned a broad NOT-TESTABLE verdict. TPC-121–131 refine the determinant, zero-mode, archive, fixed-shift, arithmetic-reduction, and end-to-end synthesis branches. We give a stricter MVP3 decision calculus with seven ordered, exhaustive route outcomes after manifest validation: GO, ARCHITECTURE-INFEASIBLE, REROUTE, STOP-ROUTE, NOT-TESTABLE, ARITHMETIC-FRONTIER, and OPEN. Contradictory route, scope, or stop records are rejected before verdict selection. The new arithmetic-frontier label is awarded only after every structural, provenance, fixed-shift, and endpoint interface is proved and every unresolved node is an explicitly typed status-OPEN, positive-L2 growing coefficient-specific arithmetic target. Reductions, finite regressions, and conditional interfaces cannot trigger it. The audited snapshot has no complete actual-carrier native archive, no complete theorem-backed endpoint registry, and no proved positive L2 estimate. Its first machine-readable missing record is H1’s native path archive, so the verdict remains NOT-TESTABLE. This is not GO, not the arithmetic frontier, and not an impossibility theorem for the architecture or for twin primes. We finish with certificate-specific branches for repair, arithmetic attack, rerouting, and publication of route obstructions.

Keywords: fixed shift; typed proof graph; route decision; arithmetic frontier; first-missing certificate; parity barrier.

Snapshot status. This audit is dated 27 July 2026 and applies to the declared TPC-MVP1 route and the TPC-121–131 certificate bundle. It proves no growing fixed- h_0 arithmetic saving and no prime-pair theorem. Failure of one sufficient route is never promoted to a negative result about the twin-prime conjecture.

1 From MVP2 to MVP3

TPC-MVP1 resolved a conditional fixed-shift proof program into gates H1–H9 [1]. TPC-MVP2 gave each gate a typed status and returned NOT-TESTABLE for the TPC-103–119 snapshot [2]. The next papers sharpen two kinds of input.

First, TPC-121 and TPC-122 replace opaque determinant and zero-mode symbols by explicit conditional exponent formulas [3, 4]. Second, TPC-123–130 specify a native path archive, physical reassembly, prescribed-shift intertwining, a literal affine block, and an exact route from determinant two to masked shift-two Liouville correlations and a Fejér four-Liouville target [5–12]. TPC-131 then states the general endpoint formula

$$|B_{h_0, \delta}(X)| \leq o(X) + X^{1-\sigma_{\text{raw}}+\Lambda_{\text{phys}}+o(1)}$$

under a proof-carrying acyclic certificate [13].

These advances make the missing inputs more precise. They do not prove those inputs. MVP3 therefore needs a state between a generic open program and a genuinely isolated arithmetic frontier, while preserving route-specific stop certificates.

2 Node status and evidence level are different

Definition 2.1 (Node status). For one fully typed node:

- PROVED: a theorem proves the declared statement;
- CONDITIONAL: an exact implication is proved but a displayed premise remains unproved;
- OPEN: the literal object and test are available but the target theorem is absent;
- NOT-TESTABLE: an archive, type, exponent, or compatibility record needed to execute the test is absent; and
- REFUTED: a theorem-backed witness contradicts that particular declared statement.

Definition 2.2 (Evidence level and direction). An L0 record is an abstract finite identity or model obstruction. An L1 record is an exact identity or executable crosswalk on a literal physical carrier. A conditional L1 record remains L1. A positive L2 record is a proved, uniform, growing, coefficient-specific power estimate at the prescribed h_0 . A negative L2 record is an actual growing counterfamily or lower obstruction in the same quantifier scope.

Finite computation over many values of X , a frozen-family estimate, a density-one result, an averaged-shift theorem, and a reduction to an unproved bilinear or four-sign form do not meet the positive L2 definition.

Proposition 2.3 (No promotion by naming). *A node labeled “arithmetic,” “spectral,” “Liouville,” “bilinear,” or “four-sign” is not thereby L2. In particular the exact pullback and decompositions of TPC-127–130 remain L0/L1 until their growing actual-family estimates are proved.*

Proof. The evidence level is determined by quantifiers, carrier, shift, normalization, and the proved inequality, not by the formula’s vocabulary. Exact transformations preserve a target but do not supply its missing bound. $\square$

3 Seven ordered exhaustive route outcomes

Definition 3.1 (Verdict certificates). For a selected literal route and a declared route universe:

- (i) GO: every required node is PROVED in one compatible scope and $\sigma_{\text{raw}} - \Lambda_{\text{phys}} > 0$.
- (ii) ARCHITECTURE-INFEASIBLE: a theorem-backed cover exhausts the declared route universe and every covered route has an unavoidable stop certificate.
- (iii) REROUTE: the selected route is stopped and a typed alternative has been selected with a fresh registry.
- (iv) STOP-ROUTE: the selected route has a complete literal stop certificate, but no typed alternative has yet been selected.
- (v) NOT-TESTABLE: a required artifact or type is absent.
- (vi) ARITHMETIC-FRONTIER: every structural, provenance, fixed-shift, and endpoint node is PROVED, strict endpoint slack is certified, and every unresolved node has status OPEN and is an explicitly typed positive-L2 target. A CONDITIONAL, REFUTED, NOT-TESTABLE, or scope-mismatched residual node blocks this outcome.
- (vii) OPEN: every remaining case.

Theorem 3.2 (Verdict precedence). *Given a snapshot manifest passing the consistency checks of definition 5.1, apply definition 3.1 in the displayed order. The first applicable test returns exactly one of the seven outcomes. An inconsistent manifest returns no route verdict and must first be repaired.*

Proof. Manifest validation first removes malformed node tables, scopes, route selections, stop covers, and reroute records. On a valid manifest, the displayed list is an ordered classifier: GO is tested first, the two global and selected-route stop branches are tested next, and the first absent artifact then gives NOT-TESTABLE. With no earlier outcome, ARITHMETIC-FRONTIER requires strict structural closure and a residual set consisting only of OPEN, positive-L2 targets; the final catch-all is OPEN. Hence some test always applies and precedence selects a unique first applicable outcome. This does not assert that the unprioritized raw predicates are intrinsically pairwise disjoint. $\square$

Remark 3.3 (Conditional synthesis is metadata). The statement “conditional synthesis available” records that a valid implication has been proved. It is not a route verdict and is not a weakened form of GO.

Remark 3.4 (Architecture-infeasible is scoped). The route universe must itself have a proof-carrying cover. Stopping all currently fashionable examples is insufficient. Even a valid ARCHITECTURE-INFEASIBLE verdict excludes only the declared architecture; it does not contradict the twin-prime conjecture.

4 Necessary certificates

Table 1: Minimum proof-carrying evidence for each positive or negative route decision.

Verdict	Required certificate	Forbidden shortcut
GO	Snapshot manifest; H1–H9 typed proof table; complete archive and reassembly; fixed- h_0 crosswalk; actual growing packet and H5 theorems; complete tail; exact H8; nonduplicated endpoint primal with strict slack.	Unknown exponent set to zero, equality at 1/400, or a conditional interface counted as a theorem.
ARCHITECTURE-INFEASIBLE	Complete route-universe cover and an unavoidable actual-carrier stop certificate on every route cell.	Failure of one frame, one archive format, or one sufficient majorant.
REROUTE	Selected-route stop witness, a typed alternative edge, and a fresh event registry for the alternative.	Carrying obsolete endpoint slack or incompatible parameter choices into the new route.
STOP-ROUTE	Actual kernel, counterfamily, incompatibility, or exact endpoint dual for the fully specified route.	Numerical failure or absence of a proof.
ARITHMETIC-FRONTIER	All nonarithmetic nodes proved, strict physical endpoint certified, and a machine check that every unresolved node has status OPEN, positive L2 evidence, and compatible scope.	Calling a reduction, finite audit, or averaged theorem the arithmetic frontier.

For NOT-TESTABLE, the certificate is a first-missing record. For OPEN, the literal object and test must be specified so that a future theorem or stop witness has fixed quantifiers.

5 Frozen snapshots and controlled changes

A route verdict is meaningful only relative to a frozen object. The word “current” must therefore be backed by data rather than by the latest paper number.

Definition 5.1 (Snapshot manifest). An MVP3 snapshot manifest contains:

- (i) the audit date and repository commit or content hashes;
- (ii) the declared route universe and the selected route;
- (iii) every node statement, scope, carrier hash, and evidence source;
- (iv) the proof-DAG edges and one topological order;
- (v) the physical occurrence registry and every allowed joint-token replacement; and
- (vi) the endpoint threshold, parameter cells, and exact certificate status.

Changing any field creates a new snapshot rather than silently updating the old verdict.

Before verdict selection, the validator requires exact agreement between the node table, the DAG node set, and the supplied topological order; checks acyclicity and parent-before-child order; validates every status, evidence tag, and scope; and requires the selected route to belong to the declared universe. Each stop record must carry a source, carrier, and scope. An ARCHITECTURE-INFEASIBLE claim must cover the route universe exactly. A reroute alternative must be distinct, declared, and not already stopped, while the selected route must be stopped. An endpoint certificate must name its source and route cell. Contradictions are rejected rather than mapped silently to a verdict.

The dated v2 manifest used here records content hashes for TPC-120 and TPC-123–132, with a joint hash for TPC-121–122; the route universe

$$\mathcal{R} = \{\text{signed_native}, \text{positive_resonance}, \text{coherent_prime_square}\};$$

the selected route `signed_native`; the explicit H1–H9 DAG and its topological order; an INCOMPLETE occurrence registry; and an endpoint record with both `proved=false` and `strict_slack=false`. A source change changes its content hash and therefore creates a new snapshot.

Proposition 5.2 (What is and is not monotone). *Fix the node statements, route universe, carrier, and physical event registry.*

- (a) *Replacing a missing artifact by a valid typed artifact can move the first-missing pointer downstream, but does not by itself prove the target at that node.*
- (b) *Adding a redundant proof to an already complete GO certificate preserves GO.*
- (c) *A route-specific stop certificate remains valid when unrelated alternatives are added.*
- (d) *An ARCHITECTURE-INFEASIBLE certificate need not survive enlargement of the declared route universe.*
- (e) *Revealing the value of a formerly unknown nonnegative endpoint token may lead either to a strict pass or to a stop; evidence completion is not numerical improvement.*

Proof. For (a), the topological scan proceeds to the next absent record, but the new artifact may have status OPEN or CONDITIONAL. Item (b) follows because neither a premise nor the strict inequality changes. A witness for one fixed route does not depend on the existence of a different route, proving (c). By contrast, ARCHITECTURE-INFEASIBLE has a universal quantifier over the declared route universe; adding an uncovered route invalidates that quantifier and proves (d). Finally, an unknown token is not zero. Its theorem-backed value can lie on either side of the strict threshold, which proves (e). $\square$

Remark 5.3 (The first missing item is not the hardest theorem). The topological first-missing report identifies the earliest point at which the present certificate cannot be executed. It does not claim that this item is analytically deepest. H3 may remain the parity-facing gate even while H1 is the first absent artifact.

Remark 5.4 (Rerouting is a new certificate). Changing from an averaged frame to direct native- h_0 dispersion, or from a positive majorant to a signed four-sign route, changes maps, scopes, and event occurrences. The old endpoint optimizer is not a warm-start proof: the alternative requires a new manifest and a new canonical registry.

6 Paper-by-paper audit: TPC-121–131

No row is a proved positive L2 theorem. The table is a dependency audit, not a statement that each proposed certificate is already instantiated on a complete growing archive.

Table 2: Strongest reusable output and the nonpromotion boundary.

TPC	Exact output	Level	Missing item
121	Phase-sector determinant certificate $(\lambda_E + 2\lambda_\phi - \gamma_R)_+$.	Proved L0 formula;	Actual growing diagonal,
122	Sharp Abel/BV transfer $\min\{\delta_{\text{pre}} - \ell_Z, \eta_{\text{cont}}\}$.	conditional L1	participation, and phase inputs.
123	Stagewise native-path conservation and first-defect format.	Proved L0/L1	Uniform actual prefixes, envelope,
124	Leaf and physical residual identities and dual tests.	transfer; growing	content, and verified TPC-111
125	Fixed-shift commutator telescope and localization cost.	premises open	reassembly.
126	Literal block extraction, exact TT^* , and prefix crosswalk.	Proved finite L0;	Complete actual-carrier growing
127	Determinant-two pullback to a masked shift-two Liouville correlation.	growing L1 schema	archive.
128	Squarefree-mask divisor/CRT expansion with explicit remainder.	Proved finite L0;	Actual matrices and original-scale
129	Quantifier firewall separating known averages from the actual target.	typed L1 interface	residual bound.
130	Exact Fejér sliding-window reduction to a shift-two four-Liouville certificate.	Proved finite L0;	Actual zero defect or subcritical
131	Acyclic typed synthesis and $\sigma_{\text{raw}} - \Lambda_{\text{phys}}$ endpoint.	conditional L1	growing Christoffel bound.
		Proved finite L0;	Uniform growing block-family power
		growing L1	saving.
		attachment	
		Proved finite L0;	Correlation estimate in the actual
		literal L1 target	progression and weights.
		Proved finite L0;	Positive relative tail exponent at
		literal L1 interface	actual N, Y, R , uniformity through
			modulus asR^4 , weighted census, and
			signed estimates on retained CRT
			fibers.
		Proved L0/L1	A theorem with growing modulus,
		firewall	origin, masks, twist, prefixes, and
			ordinary weight.
		Proved finite L0;	Uniform actual Fejér four-Liouville
		literal L1 target	estimate with participation, masks,
			phases, weights, origins, and prefixes
			over the complete family.
			Every actual primitive theorem and
			complete physical registry.

Table 3: MVP3 snapshot gate status on 27 July 2026.

Gate	Status	Evidence target	First unresolved record
H1	NOT-TESTABLE	L1	Complete actual-carrier native path archive.
H2	OPEN	positive L2	Literal growing resonance ledgers or a signed replacement.
H3	OPEN	positive L2	Uniform actual Fejér four-Liouville estimate with
H4	OPEN	positive L2	participation, masks, phases, weights, origins, and prefixes.
H5	NOT-TESTABLE	positive L2	Complete original-scale physical tail estimate.
H6	NOT-TESTABLE	L1	Actual compatible λ_D, η_Z certificate pair.
H7	NOT-TESTABLE	L1	Actual growing physical cover or residual certificate.
H8	NOT-TESTABLE	L1	Complete shift-tagged growing archive or subcritical
H9	NOT-TESTABLE	L1	localization certificate.
			Complete exact hard-packet intertwining.
			Complete theorem-backed event registry and strict primal
			certificate.

7 Projection onto H1–H9

Definition 7.1 (First-missing report). The machine manifest stores an explicit gate-certificate DAG and a candidate topological order. It first rejects cycles, missing or extra node IDs, and any order that fails exact node coverage or parent-before-child precedence. It then scans the validated order. The first-missing report is

(node, gate, status, evidence, required artifact)

for the earliest required node having status NOT-TESTABLE.

The v2 DAG makes H1 the root; H2–H7 depend on H1; H8 depends on H1, H6, and H7; and H9 depends on H2–H8. Its validated machine scan therefore returns

node = H1.archive,
artifact = complete actual-carrier native path archive.

The report is stable under downstream speculation: a later conditional theorem cannot make this earlier archive executable.

8 MVP3 snapshot verdict

Theorem 8.1 (MVP3 decision). *For the declared TPC-MVP1 architecture and the TPC-121–131 certificate bundle audited on 27 July 2026,*

Verdict_{121:131} = NOT-TESTABLE.

The snapshot is neither GO, nor ARITHMETIC-FRONTIER, nor ARCHITECTURE-INFEASIBLE.

Proof. H1’s complete actual-carrier archive is absent, so [theorem 3.2](#) reaches NOT-TESTABLE before either ARITHMETIC-FRONTIER or OPEN. The GO premise fails additionally at the open H2–H4 arithmetic and tail targets and the not-testable H5–H9 records. The arithmetic-frontier premise fails because several structural nodes are not proved and no strict actual endpoint is certified. Finally, the stop records cover only `positive_resonance` and `coherent_prime_square`; the selected `signed_native` route has no stop certificate. Hence no exact cover of the declared route universe exists and ARCHITECTURE-INFEASIBLE is unavailable. $\square$

Corollary 8.2 (No positive L2 progress claim). *The TPC-121–131 bundle may sharpen arithmetic targets and exact crosswalks, but it contains no theorem that can be entered as a positive L2 gate in [table 3](#).*

Proof. Each row of the paper audit either proves a finite identity, conditional implication, reduction, or quantifier obstruction, or states a growing estimate as the missing item. None proves the required uniform coefficient-specific power estimate. $\square$

9 Certified next-stage branching

On a reroute, determinant reserve, old parameter optima, and joint tokens do not carry over automatically. The alternative begins with a new scope record and occurrence registry.

10 Machine regression and claim boundary

The standard-library script `experiments/tpc132_mv3_route_audit.py` encodes the gate records and verifies:

1. reachability and precedence of all seven ordered outcomes after manifest validation;

Table 4: Response determined by the first valid certificate.

Observed certificate	Next branch	What may be published
Missing archive or path metadata	Build only the first missing stage; preserve atom keys, h_0 , normalization, order, and all tail namespaces.	Executable archive schema and a dated not-testable audit.
Leaf/physical residual witness	Repair the grouping or prove its literal physical pairing $o(X)$; do not infer one residual from the other.	Route-specific reassembly obstruction.
Fixed- h_0 range or commutator failure	Abandon the averaged/frozen transfer and move to a direct native h_0 theorem.	An averaged-only theorem with the correct qualifier.
All structural nodes proved; only L2 targets remain	Freeze the interfaces and attack the actual H3 Fejér four-Liouville estimate, H5, H4, or a signed-H2 theorem.	An honest arithmetic-frontier statement.
$\lambda_D > 2\eta_Z$ on actual inputs	Improve phase participation, signed prefixes/content, or replace H5 by a direct determinant theorem.	Compatibility stop for the selected H5 mechanism.
$\Lambda_{\text{phys}} \geq 1/400$ by exact dual	Stop that registry; choose a new frame/composite or direct native route with a fresh ledger.	Strict endpoint obstruction.
Fejér four-Liouville target proved equivalent to an unavailable stronger theorem	Publish the no-reduction boundary and change the arithmetic decomposition.	Hardness/equivalence result, not parity progress.

2. rejection of arithmetic-frontier labels with a conditional or refuted residual node;
3. refusal to return ARCHITECTURE-INFEASIBLE without a complete route-universe certificate and rejection of contradictory manifests;
4. presence of the content-hash manifest, explicit DAG, and valid topological order;
5. absence of a proved positive-L2 record; and
6. the DAG-checked machine-readable first-missing report for H1.

The script is a decision-logic regression. It does not inspect primes or establish any growing analytic estimate.

Established. The verdict precedence, certificate requirements, first-missing format, and snapshot classification are L0/L1 audit results. They prevent circular implication, false route closure, and false promotion to the arithmetic frontier.

Not established. No complete actual archive, H1–H9 conjunction, positive fixed-shift L2 theorem, strict physical endpoint, hard-packet $o(X)$ estimate, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem is proved.

11 Conclusion

MVP3 turns “we are near the arithmetic core” into a checkable statement rather than a rhetorical one. The current program has a sharper map and several exact transformations, but its first missing object is still structural. The correct verdict is therefore NOT-TESTABLE, not ARITHMETIC-FRONTIER. If the archive, reassembly, fixed-shift, and endpoint certificates are eventually closed, the same decision calculus will promote the route to the arithmetic frontier only when every remaining unresolved node is OPEN, positive L2, and scope-compatible, without calling a conditional interface or reduction a theorem. If a literal stop appears first, the calculus records the stopped route and the permitted alternative without implying anything negative about twin primes.

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